

Lab 2 - Linguistics Data, STAT 215A, Fall 2026

October 3, 2026

1 Introduction

Ask two people from the same city how they address a group of strangers, or what they call a fizzy drink, and there is a good chance they will not answer alike, but ask the same question of two people three hundred miles apart and the odds of agreement drop sharply. American English varies systematically with geography, and Bert Vaux’s Dialect Survey was built to measure exactly that: 47,471 respondents answering the same lexical questions, geolocated by the ZIP codes they reported.

Measuring the geography is the easy part; interpreting it is not. Any single word’s distribution is a tangle of local pockets, migration, and contact, so dialectometry aggregates many variables at once. Nerbonne and Kretzschmar’s 2003 introduction frames dialectometry as the computational counterpart of dialectology, the measurement of variation whose distribution is primarily geographic, and their 2006 follow-up argues that the payoff comes from explaining aggregate patterns rather than individual words. That is the logic here: two words alone, then four, then all 67, and finally a clustering of every respondent, with each step asking whether the structure that emerges is geography showing through or an artifact of the encoding.

2 The Data

Bert Vaux’s Dialect Survey put roughly one hundred questions to 47,471 respondents across the United States, and this analysis uses three files from it. `lingData.txt` is the respondent-level data: one row per person, 73 columns wide, holding an anonymous ID, the city, state and ZIP code each respondent typed in, their answers to the 67 lexical questions in the range Q050–Q121, and a latitude and longitude derived from their ZIP code. Almost everyone answered almost everything, so the interesting question is what people *say* rather than whether they answered.

The answers arrive as integers rather than words. A 0 means the question was not answered; any other number indexes the survey’s published answer letters, so a 1 means the respondent picked option (a). `question_data.RData` supplies the question wording and the answer letters for every item, so Q50 can be read as “what word do you use to address a group of two or more people?” and its code 4 as *you guys*. This analysis loads that file with `pyreadr` to turn codes into labels wherever answers are named in text or figures.

The third file, `lingLocation.txt`, holds no new respondents. It is the same answers re-encoded geographically: each person’s answers become a binary indicator vector, and everyone sharing a one-degree square of latitude and longitude is *summed* into one row for that cell. A cell read 50 for “soda” therefore means 50 people in that square said soda, not that 50 is a proportion. Only Section 5 uses these cells, and Section 5.4 shows why that turns out to be the fragile part of the analysis; everything else works at the respondent level. Map overlays follow the pattern in `code/demo.ipynb`.

2.1 Data Cleaning

The data are messier than they look, and the mess is worth naming because several cleaning choices are judgment calls that affect later results.

No-response codes. 3.17% of all answer cells are 0, with per-question response rates from 91.8% (Q092, lane-changing) to 97.3% (Q050, Q052, Q054, Q056, Q063). A 0 conflates two different situations: respondents who skipped the question, and respondents who genuinely have no word for the concept. Q064 offers an explicit “I have no word for this” (code 10, 0.9% of respondents), so a 0 is not the same as a shrug. I keep 0s as zeros through the binary encoding rather than dropping rows, since dropping would discard geography:

respondents who skipped Q50 still answered Q105 and still have a location.

State codes. STATE has 92 distinct values, of which 159 rows are not US states: XX placeholders (84 rows), Canadian provinces (ON, NB, BC, QC, NS, AB), UK/NZ/IE/PR, and outright typos (00, 94, C), !L). These rows are flagged via `state_valid` rather than dropped. At 159 of 47,471 they cannot move any regional statistic, and dropping them silently would hide a real data-entry problem worth reporting.

Geocoding. 1,020 respondents have no coordinates, leaving 46,245 mappable; 206 more fall outside a continental bounding box because the ZIP centroids fall in Alaska, Hawaii, or outside the country. Because coordinates come from ZIP centroids rather than respondent-reported positions, they are only as precise as the ZIP code area, which is fine for regional patterns and poor for anything finer. All of this lives in `code/clean.py`, whose `clean_data` copies its input and returns a new frame with `state_valid` and `geo_valid` flags, so each decision above is recorded once and applied everywhere downstream.

3 Exploratory Data Analysis

3.1 Two questions and their geography

I focus on Q50 (what word do you use to address a group of two or more people?) and Q105 (your general term for a sweetened carbonated beverage). Among the 46,205 respondents who answered Q50, the split is *you guys* 42%, *you* 24%, *y'all* 18%, *you all* 13%. Among the 46,098 who answered Q105 it is *soda* 48%, *pop* 28%, *coke* 14%. Restricted instead to respondents giving one of the three standard beverage words, the Q105 split is 53/31/16 and Q50 is 44/25/18/13. Figure 1 shows both mapped, and the two answer different questions about the country. Grouping the respondents into four Census-style regions, defined as in Table 1 (the Census South, so Delaware, Maryland and DC are counted as South), shows how sharply: *coke* reaches 41.4% in the South and under 10% everywhere else, *soda* takes 81.3% of the Northeast, and *pop* leads the Midwest at 55.2%. The West is the interesting case: majority *soda* at 55.1% but a substantial *pop* minority at 33.8%, which is why the country is not cleanly split in two.

Region	<i>soda</i>	<i>pop</i>	<i>coke</i>
Northeast	81.3	13.2	5.5
Midwest	35.6	55.2	9.2
South	46.8	11.7	41.4
West	55.1	33.8	11.0

Table 1: Q105 answers by region (percent of respondents in that region giving each answer). Regions follow the Census (South includes DE, MD and DC); 1,020 respondents without usable coordinates and 159 with unusable state codes are excluded, leaving 46,292. Sample shares: Midwest 34.5%, Northeast 24.6%, South 22.6%, West 18.3%.

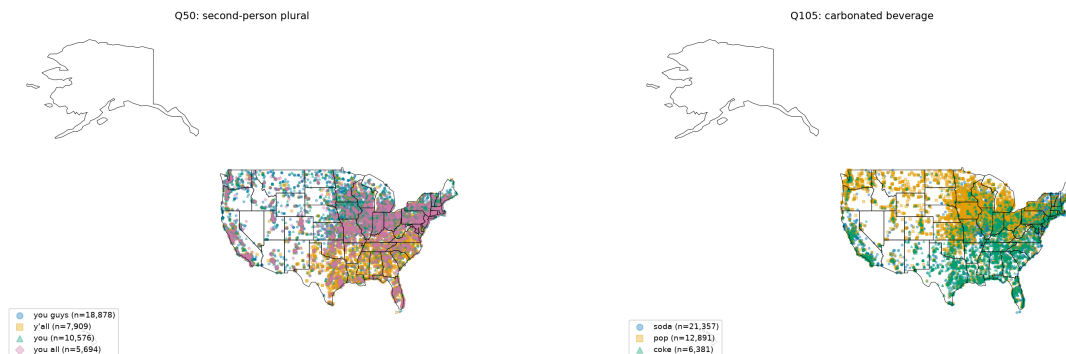


Figure 1: Q50 (left) and Q105 (right) answers by respondent location.

The survey's own published percentages, carried in `question_data.RData`, are a useful external check: our subset closely reproduces them on the standard-answer denominator (Q105 *soda* 53.0% reported against

52.9% here; Q50 *you* 24.8% against 24.6%). The substantive differences are *pop* (25.1% reported, 31.4% here) and *y'all* (14.0% against 18.3%), so this PUD subset skews slightly Southern in exactly the direction of the effects I report.

Q64 and Q103 are geographically precise in a different way. Q64 (*sub* / *hoagie* / *hero* / *grinder*) is the standard textbook illustration of an enclave rather than a gradient: *hoagie* is given by 15.7% of respondents in the Northeast against 3.9% everywhere else. Q103 is the reverse case: *bubbler* (water fountain in schools) is given by 33.8% of respondents in WI/MA/RI against 1.3% elsewhere, a genuinely local word with a tiny footprint. Q73 (*sneakers* vs *tennis shoes*) leans Southern too, though only mildly (56.6% vs 44.0% for *tennis shoes*).

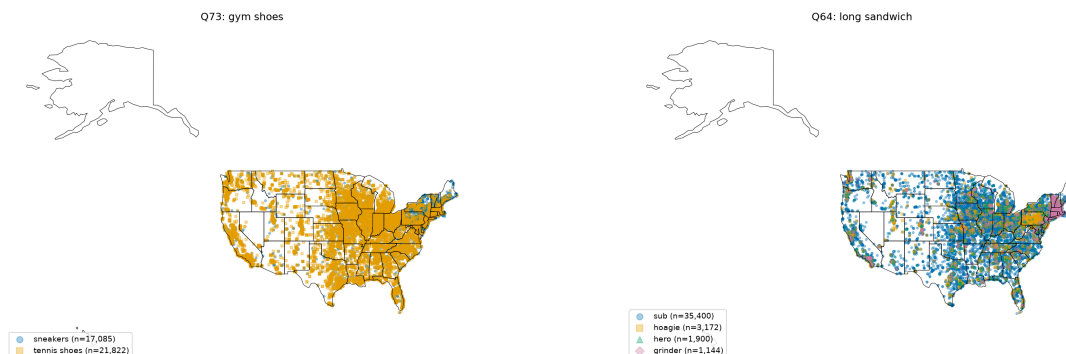


Figure 2: Q73 (left): *tennis shoes* skew Southern. Q64 (right): the Philadelphia *hoagie* pocket.

The scatter maps overplot tens of thousands of points, so density is hard to read. Figure 3 replaces them with a KNN-smoothed probability surface (100 nearest neighbours in lon/lat, distance-weighted), which answers the region-versus-gradient question directly: *coke* never exceeds $P = 0.85$ anywhere and holds a majority over only 13% of the grid, *pop* covers 27%, and *soda* 47%. Three broad surfaces with soft, overlapping edges is itself evidence for the continuum reading in Section 5.3.

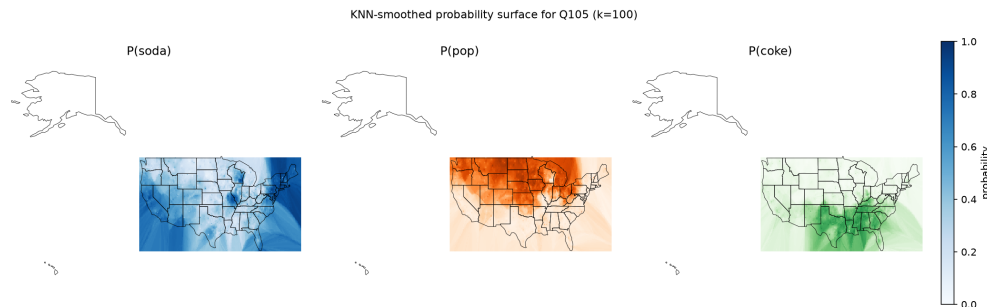


Figure 3: KNN-smoothed probability of each Q105 answer across the continental US. The three surfaces meet in soft transitions rather than sharp borders.

Because static maps hide which words co-occur in the same mouths, I also built an interactive version (Figure 4); hovering any point reports that respondent's Q50 answer alongside their map color, which makes the *coke*/*y'all* overlap in the Deep South directly visible.

3.2 Does one answer help predict the other?

Screening all 2,211 question pairs first (Section 5.4) shows that Q50×Q105 is well above the median pair association but is not the most strongly associated pair in the survey: kinship terms (Q070×Q071, $V = 0.414$) and the *anymore* sentences (Q054×Q055, $V = 0.417$) are stronger. Those pairs are near-deterministic within families, whereas Q50 and Q105 were picked for geographic reach rather than raw association, which is the trade I wanted.

Q105 beverage word by respondent location (hover: Q50 answer)

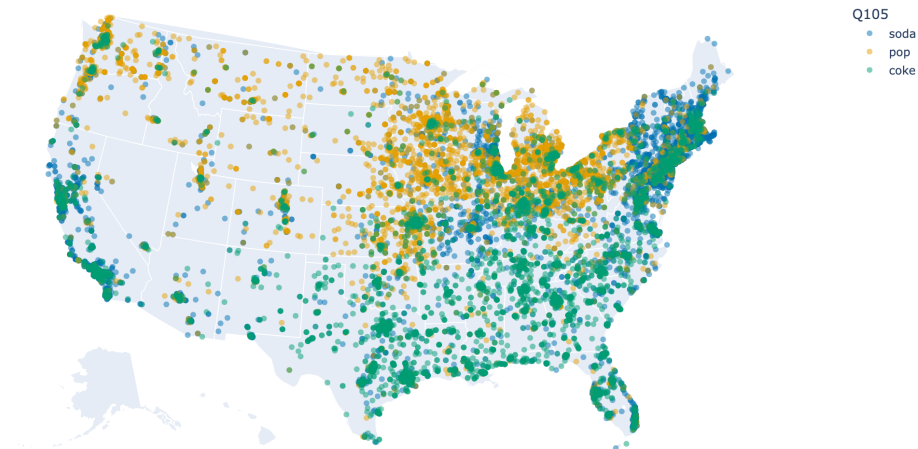


Figure 4: Q105 by location, exported from the interactive Plotly version ([figs/interactive_q105.html](#), bundled for offline hover/zoom). The interactive version plots the 24,579 respondents who gave *soda*, *pop*, or *coke* and answered Q50 with *you guys* or *y'all*.

The two questions are not independent. Among respondents answering both, $P(y'all \mid coke) = 58.3\%$, against 6.0% given *pop* and 12.2% given *soda*, roughly a tenfold lift. Cramér’s V over the full four-by-three table is 0.276 ($n = 42,320$). The association is asymmetric: knowing someone says *coke* says a great deal about their pronoun, whereas knowing *y'all* says much less about their drink, because *you guys* is the majority answer nearly everywhere and so carries little information about the beverage word.

Table 2 shows the joint distribution behind these numbers. The *coke* column holds 3,684 *y'all* respondents while the *pop* column holds only 750, a nearly fivefold difference that is almost entirely a *y'all*-vs-*you guys* split: *you guys* is 57.3% of the *pop* column but only 23.1% of the *coke* column.

	<i>soda</i>	<i>pop</i>	<i>coke</i>	other
<i>you all</i>	2,775	1,647	673	477
<i>you guys</i>	9,577	7,140	1,460	543
<i>you</i>	6,076	2,928	500	873
<i>y'all</i>	2,560	750	3,684	657

Table 2: Q50 rows by Q105 columns (respondents answering both, $n = 42,320$). Cramér’s V = 0.276.

Turning that into an actual prediction makes the asymmetry precise. Among the respondents who gave one of the three beverage words, 15.7% say *coke*. Predicting that binary outcome from Q50 alone with a one-hot logistic model reaches 85.2% accuracy under five-fold cross-validation, 5.4 times the base rate’s error rate, and adding Q73 raises it to 87.6%. A single lexical question therefore carries substantial information about an unrelated one, which is precisely the aggregation argument Nerbonne and Kretzschmar make: individual variables are noisy, but their joint geography is not.

3.3 Beyond two questions

Extending to four questions shows one dominant axis and several local ones. The pairwise associations rank Q50×Q105 first ($V = 0.276$), then Q105×Q73 (0.187), Q105×Q64 (0.164), Q73×Q64 (0.145), with both Q50 cross-questions weakest (0.077 and 0.082). The pattern is what dialectometry predicts: a broad north–south axis that many words share, plus sharp local enclaves (*hoagie*, *bubbler*) that participate in almost no

large-scale structure. Answer codes used throughout are listed in Table 3.

Question	Codes used	Answers
Q50	1, 4, 7, 9	<i>you all, you guys, you, y'all</i>
Q105	1, 2, 3	<i>soda, pop, coke</i>
Q73	1, 6	<i>sneakers, tennis shoes</i>
Q64	1, 3, 4, 2	<i>sub, hoagie, hero, grinder</i>
Q103	1, 4	<i>bubbler, water fountain</i>

Table 3: Question codes used in this report, as returned by `question_data.RData`.

4 Dimension Reduction

4.1 Encoding

Encoding every (question, answer) pair as a binary indicator turns 67 categorical questions into a $(47,471 \times 468)$ binary matrix at 13.9% density, the $p = 468$ given in the instructions. Answers like Q065’s “I use lightning bug and firefly interchangeably” become their own column alongside the two individual answers, and a 0 (no response) contributes zeros across that question’s whole block.

4.2 Centering and scaling, and why not PCA on the raw categories

Two choices matter here, and the instructions ask for both to be defended. *Centering* is handled internally by scikit-learn’s PCA, so uncentered and centered fits are identical; I report both to make that explicit. On the resulting components, PC1 explains 4.03% of variance and PC2 3.46%, with 10 components covering 23.15%: a flat spectrum, which for 468 sparse dimensions is expected and itself informative: there is no small set of dominant directions, because each component is a diffuse regional mixture.

Scaling is where I deviate from the default. Standardizing every column to unit variance roughly halves the concentration (PC1 1.90%, ten components 11.65% against 23.15%). That is a bad trade here: all 468 columns are already 0/1 with identical scale, so standardizing cannot correct any real units mismatch. It can only reweight, promoting answers given by a few hundred respondents (rare *other* categories) to parity with answers given by tens of thousands. Frequency is genuine linguistic signal, not a nuisance parameter. So I neither center nor scale.

The projections also look different from each other, which the instructions ask about directly. Figure 5 puts uncentered, centered, and standardized side by side: the centered and uncentered panels are indistinguishable, while the standardized panel visibly redistributes respondents along both axes, compressing the coastal group toward the middle. Choosing a projection is therefore not a neutral preprocessing step; it reshapes what the data appear to say.

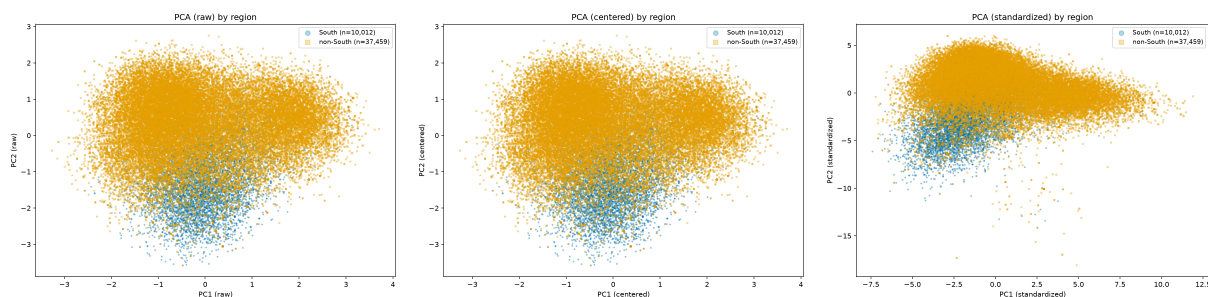


Figure 5: PCA projections of the same respondents, colored by region: uncentered, centered, and standardized. The first two are identical; standardization redistributes the coastal group.

Running PCA directly on `lingData` instead of the encoded matrix would be actively misleading. The raw codes are labels, not measurements: the gap between code 4 (*you guys*) and code 9 (*y'all*) is 5, while the gap between code 4 and code 7 (*you*) is 3, yet these three words are equally different ways of addressing a group.

Euclidean distance in code space would therefore report that *you-guys* is "closer to" *you* than to *y'all*, and PCA's variance-maximizing directions would be built on meaningless arithmetic. One-hot encoding removes the ordering that was never there.

4.3 NMF and t-SNE

PCA assumes linearity and roughly Gaussian variation, which fits count data poorly. NMF suits binary indicators better because it is constrained to nonnegative, additive parts. With ten components the parts read as dialect profiles rather than abstract axes (Figure 6): one bundles *sneakers* with *soda* (coastal/urban), another *tennis shoes* with *crawdad* and *lightning bug* (Southern), another double-*no* grammar answers with plain *you* (Northern/Midland). That these cohere into recognizable bundles is the clearest evidence in the analysis that the regional signal is real and not an artifact of any one question.

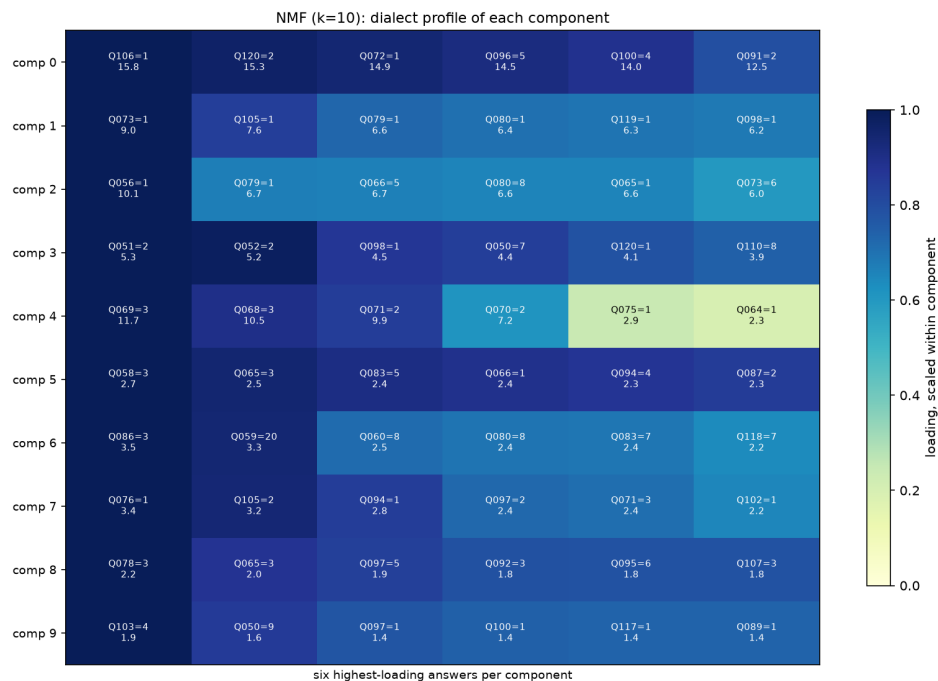


Figure 6: NMF with ten components, each showing its six highest-loading answers. Colour is scaled within a component, since components differ in total loading by almost a factor of ten. Several read as coherent regional bundles: component 1 is *sneakers* with *soda*, component 2 pairs *tennis shoes* with *crawdad* and *lightning bug*, and component 4 is the kinship set Q068–Q071.

t-SNE on PCA-50 scores (Figure 7) gives the sharpest visual separation: a distinct Southern cloud with the northern respondents forming a connected mass behind it, and very few points suspended between. Perplexity is the one parameter worth setting deliberately: it trades local detail against global structure. I used 50, which preserves the Northern/Midland sub-structure visible in PCA. I did not sweep it, so a larger value might separate the regions more cleanly or merge them further, and I cannot say which.

5 Clustering

5.1 Choosing the number of clusters

Rather than picking k for interpretability first, I swept it. Inertia falls smoothly with no elbow (1,657,134 at $k = 2$ down to 1,524,474 at $k = 8$), and silhouette scores computed on an 8,000-respondent subsample are remarkably flat: 0.033 at $k = 3$, 0.032 at $k = 2$ and $k = 5$, 0.031 at $k = 4$, falling to 0.029 by $k = 6$. These are very low values in absolute terms: under conventional thresholds, the data contain no well-separated groups at any k . What $k = 3$ does have is the highest silhouette of the sweep, and the cluster profiles are

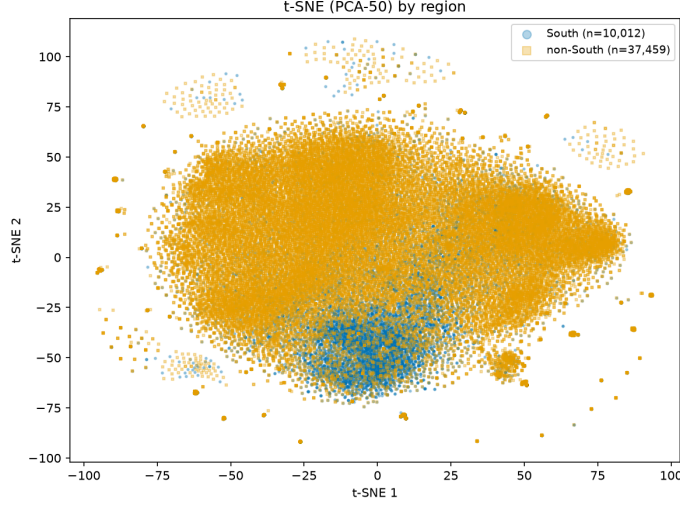


Figure 7: t-SNE on PCA-50 scores (perplexity 50): a distinct Southern cloud with the northern mass connected behind it.

interpretable enough to corroborate the choice on their own.

5.2 k-means on respondents

Running k-means on the full respondent matrix produces clusters whose modal answers are immediately interpretable. Sizes are 13,631 / 13,531 / 20,309, and each cluster has answers that are over-represented within it (Figure 8). Cluster 0 is *sneakers* (88.9% against 37.2% overall) and *soda* (84.4% vs 46.4%). Cluster 1 is *y'all* (47.3% vs 17.1%) and *coke* (37.1% vs 13.8%). Cluster 2 is *tp'ing* (82.6% vs 58.1%), *kitty-corner* (71.4% vs 45.9%), and *pop* (51.1% vs 27.5%). Mapping them (Figure 9) and cross-tabulating against the four regions (Table 4) gives Coastal/Northeast, South, and a third group that is *pop*-leaning but geographically split between the Midwest and the West, at 65.0% and 73.9% of those two regions.

Region	Cluster 0	Cluster 1	Cluster 2
Northeast	73.2	13.8	13.0
South	20.4	62.5	17.0
Midwest	10.9	24.1	65.0
West	13.3	12.7	73.9

Table 4: Cluster membership by region (row percentages). Cluster 0 = Coastal/Northeast, cluster 1 = South, cluster 2 = *pop*-leaning interior and Pacific. Cramér's V = 0.512 (n = 47,312).

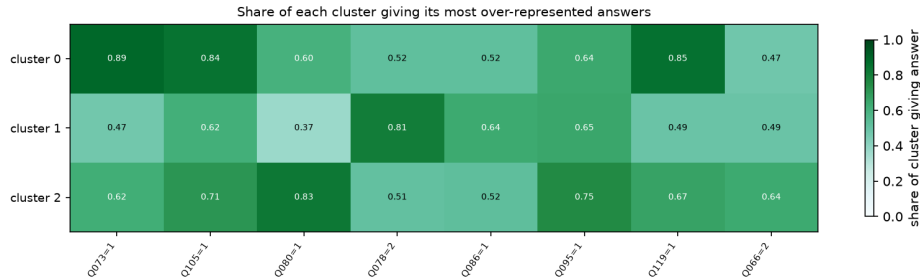


Figure 8: The eight answers most over-represented in each k-means cluster, as a share of that cluster.

The $k = 4$ solution is instructive rather than useful: it splits off a cluster of 1,395 respondents whose modal

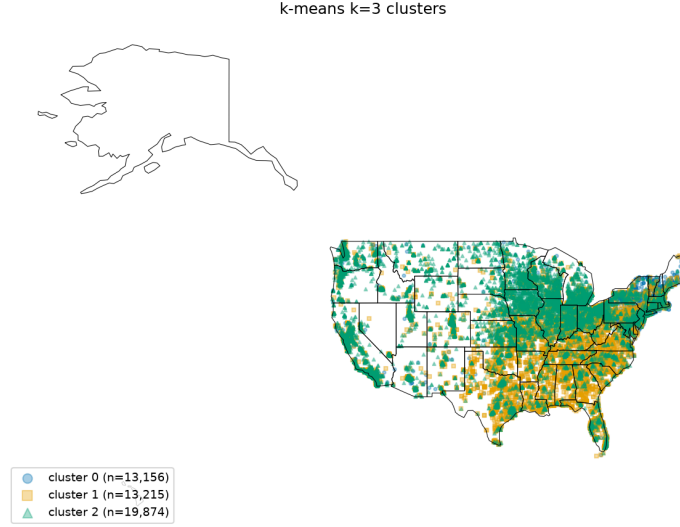


Figure 9: k-means ($k = 3$) clusters mapped: a Northeast/coastal cluster, a Southern cluster, and a *pop*-leaning interior and Pacific cluster.

answers are 0 on both focal questions: non-responders. Because my encoding maps no-response to zeros, a missingness pattern becomes a geometrically coherent group, which is exactly the failure mode one should worry about. Rejecting $k = 4$ on these grounds independently of the silhouette argument is reassuring rather than convenient.

5.3 Continuum or clusters, and from where to where?

The clusters overlap (silhouette ≈ 0.03), and the gradient figure shows why: the Southern cluster is not a discrete blob but the high end of a north-south gradient. Binning respondents into 4-degree latitude bands (finer bins in Figure 10), $P(y'all)$ runs 29%, 54%, 36%, 18%, 8%, 7% from the Gulf Coast north to the Canadian border, and $P(coke)$ tracks it at 43% in the Deep South band against 4% in the north. Longitude tells a complementary story. In 10-degree bands, $P(y'all)$ peaks at 34% in the -105 to -95 band (Missouri, Arkansas, eastern Texas), against 8% on the Pacific coast and 9% in the Mountain West. It rises again to 21% in the -85 to -75 band, which reflects Appalachian and Piedmont populations. So the continuum runs from the Pacific Coast and Mountain West (soda, *you guys*), through the Midwest (*pop*), across the Mississippi Valley, to the Deep South and Appalachia (*coke*, *y'all*), with sharp enclaves superimposed rather than replacing it.

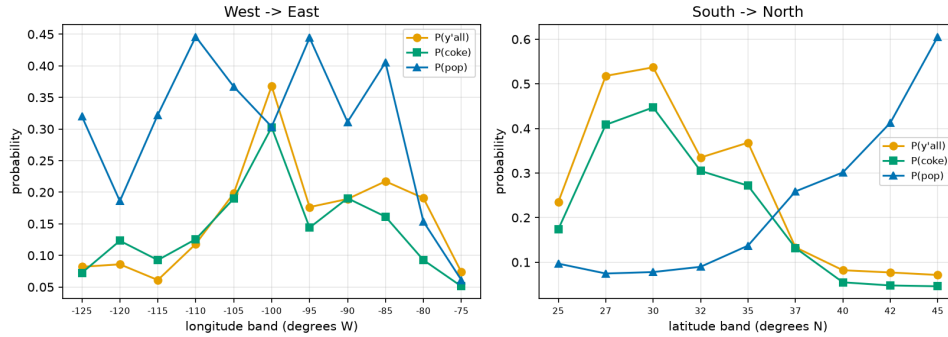


Figure 10: $P(y'all)$ and $P(coke)$ by longitude (left) and latitude (right): the southern cluster is the high end of a gradient, not an island.

5.4 Which questions produce the structure?

Ranking every question by Cramér’s V against the $k = 3$ labels identifies what drives the partition: Q105 (0.498), Q073 (0.490), Q106 (0.431), Q050 (0.418), Q103 (0.398), Q076 (0.388), Q080 (0.365), Q099 (0.343). At the bottom, Q057 (0.040) and Q121 (0.066) are essentially uninformative about region. Table 5 lists the extremes. The broad north–south variables sit at the top (Q105, Q073, Q050), and Q106 (*tp’ing* vs *toilet papering*) ranks third, which is a reminder that the strongest separators are not automatically the most interesting dialect variables: *tp’ing* is a broad informal-register marker that happens to split the country nearly the way *y’all* does. The genuinely weak separators are questions whose answers are near-universal nationwide, so they cannot distinguish anything.

Strongest separators		Weakest separators	
Q105 beverage word	0.498	Q115 academic point	0.080
Q73 gym shoes	0.490	Q121 ogling	0.066
Q106 <i>tp’ing</i>	0.431	Q057 <i>anymore</i>	0.040
Q50 group address	0.418		
Q103 water fountain	0.398		

Table 5: Cramér’s V between each question and the $k = 3$ cluster labels, highest and lowest.

5.5 Hierarchical clustering on binned cells

As a second method I cluster the 781 binned cells of `lingLocation.txt`, which directly tests whether geography-based aggregation changes the picture. The linkage choice dominates the result. Ward on row-normalized compositions isolates a single 1-person cell as its own cluster (sizes 318 / 1 / 462 across all 781 cells), because Ward merges on Euclidean distance and that lone cell sits far from everything. Average linkage collapses even harder (779 / 1 / 1). Only complete linkage on the 449 cells with at least 10 respondents gives usable groups (247 / 154 / 48), and it agrees geographically with k-means’ South-versus-rest split.

That agreement does not survive a change of metric. Jaccard distance, the natural similarity for binary indicator vectors because the thousands of questions both cells left blank cancel out, puts 444 of the 449 cells in a single group and agrees with the Euclidean partition at $\text{ARI} = 0.001$. The reason is visible in the distances themselves: every cell answered 285 of the 468 questions on average, so any two cells agree on most of what they answered and the pairwise Jaccard distances are small and nearly uniform (median 0.326, standard deviation 0.082, range 0.054–0.630). With so little to separate on, average linkage collapses.

The deeper problem is that `lingLocation.txt` rows are unnormalized counts, so a cell with many respondents shares indicators with every other large cell regardless of what was said. Jaccard on this representation partly measures survey effort rather than speech. The cell-level result is therefore fragile and metric-dependent, and I would not defend it; the respondent-level k-means result is the one that holds up.

5.6 Do the models’ assumptions fit the data?

This is the question the instructions pose about dimension reduction, and the answer differs by method. PCA’s model (orthogonal axes of maximum variance, implied by a linear-Gaussian picture) fits the *aggregate* structure well enough to expose the regional gradient, but its flat spectrum and Gaussian assumption are poor descriptions of 468 sparse indicators; NMF’s nonnegative, additive parts describe them better and yield the most interpretable components. For clustering, k-means assumes compact spherical clusters under Euclidean distance, and the silhouette of ≈ 0.03 says that assumption is violated: these clusters are not spheres, they are overlapping regional densities. Complete-linkage agglomerative clustering makes no distributional assumption at all, which is why it is the more defensible of the two here, but it assumes every point is worth as much as every other, which is how the 1-person cell ended up isolated. Neither method’s assumptions match the data perfectly; that both nonetheless recover the same geography is evidence the signal is robust rather than an artifact of one algorithm. The practical advantages divide cleanly. k-means scales to 47,471 rows in seconds and needs only k , but it requires a guess at k and can fall into poor local optima (Section 6). Complete linkage makes no shape assumption and gives a dendrogram showing structure at every level, and it is deterministic, but it costs $O(n^2)$ memory and was infeasible at full respondent count.

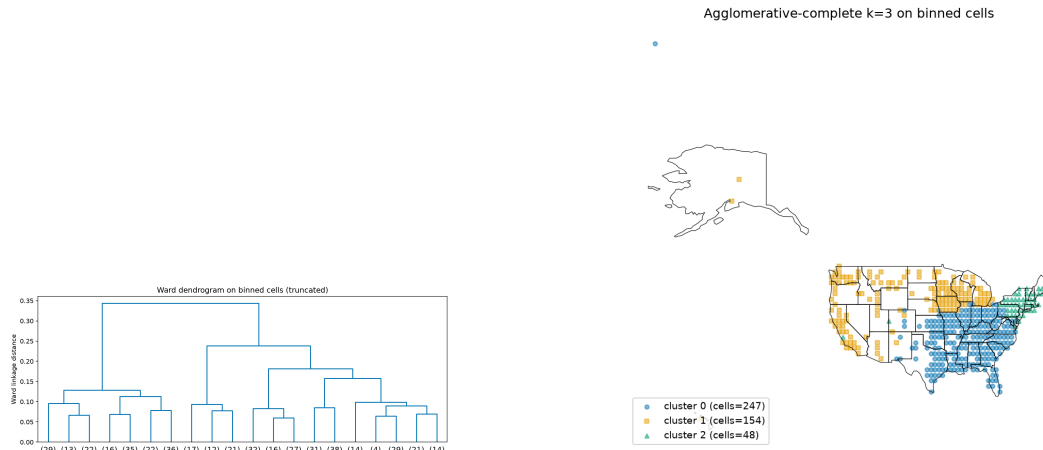


Figure 11: Ward dendrogram on the same 449 adequately populated cells used for the complete-linkage fit (left), and those complete-linkage clusters ($k = 3$, right). The degenerate Ward solution discussed in the text is computed on all 781 cells.

6 Stability of findings to perturbation

Stability is what separates a regional pattern from an artifact of one run. Comparing each refit against the full-data $k = 3$ partition by Adjusted Rand Index:

- **Different starting points.** Across 10 seeds, mean ARI = 0.944 (minimum 0.803). One seed in ten finds a visibly worse local optimum, which is the concrete argument for always reporting $n_{\text{init}} \geq 10$ rather than a single fit.
- **Subsampling.** Refitting on ten random 80% subsamples gives ARI 0.972 ± 0.019 and NMI 0.951 ± 0.028 , so the partition is essentially unchanged when a fifth of the respondents are removed, and the modest standard deviation tracks the southern cluster’s boundary, which is where respondents sit closest to the cluster edges.

A third check asks whether the *projection* generalizes, not just the partition. Fitting PCA on 70% of respondents, projecting the held-out 30% onto those components, then refitting on the held-out data alone, gives $R^2 = 0.998, 0.997$ and 0.998 on the first three components and 0.971 across ten, with matching-component cosines of 0.92 – 0.998 . PC1 on the unseen respondents still separates the South ($+0.054$) from the non-South (-0.023), so the regional axis is a property of the variables rather than of the rows used to find it.

(Bootstrapping with replacement is the obvious third check, but duplicate row indices make a point-wise partition comparison ill-defined, so subsampling is the cleaner stability read here.) The conclusion is that the Southern cluster is not an artifact of particular rows or a lucky initialization: it survives both perturbation types with ARI above 0.94, and the geographic gradient in Figure 10, a smooth monotonic band rather than a fitted boundary, is the substantive reason to trust it.

7 Conclusion

7.1 Three realms

Data. This survey describes broad regional structure well: the north–south gradient, the Q50/Q105 interaction, and the local enclaves are all large, consistent, and visible without modeling. But it is a self-selected web sample (young, online, English-literate, and not drawn from any probability frame) with self-reported locations and ZIP-centroid geocoding, so it supports description, not inference about individuals or small areas.

Algorithms. The clusters are stable (ARI > 0.94) but barely separated (silhouette ≈ 0.03). That combi-

nation says something precise: the algorithms have found the *density modes* of a continuum, not discrete populations. Reading cluster boundaries as dialect borders would misread the result: respondents form overlapping regional dialects, and the three-cluster solution is a convenient description of three places where those dialects are densest.

Future data. For decisions about people or places, this sample is not adequate without a probability-based design and survey weighting, and the geocoding precision is a hard floor on sub-county claims. What the analysis does support is that region predicts a bundle of lexical choices strongly enough to be useful for coarse planning where it already holds (language model training data, speech-recognition dialect coverage, or targeting), and the Q105 holdout result raises confidence in the southern cluster specifically.

7.2 Reality check

The clusters deserve an external test rather than internal consistency, so I removed the beverage question from the analysis entirely and re-clustered on the remaining 458 columns. If the southern cluster is a real regional phenomenon, then respondents who actually said *coke* should land in the *y'all*-heavy cluster at a much higher rate than chance, without the beverage question having contributed to that cluster's existence. They do: the *y'all*-heavy cluster (45.5% *y'all* against 4.8% and 5.4% in the other two) captures 72.1% of the 6,550 *coke* respondents, versus 19% of *pop* respondents and 17% of *soda* respondents, a fourfold enrichment. Part of this signal flows through Q050 itself, since *y'all* and *coke* are themselves correlated, so the check validates the cluster's geography more than it replicates the beverage finding independently. It is still informative: the clustering never had access to any beverage information.

7.3 Impact and what I would add next

The domain-level conclusion is that the large dialect features the literature describes are all present here (a northern/southern lexical divide, a Midland *pop* belt, sharp metropolitan enclaves), and that the aggregate-measurement approach Nerbonne and Kretzschmar advocate does recover them from noisy individual words. On the survey's own website the words I clustered on are collected precisely because they are expected to be geographically diagnostic, and they behave that way.

With more time I would add several things. Aggregating to counties would let me test whether the enclaves (*hoagie*, *bubbler*) sharpen or dissolve once the population base is large enough to estimate them reliably, since clustering the current data necessarily down-weights exactly those rare answers. Surveying age and year of response would allow the more interesting question of whether these dialects are stable or undergoing convergence. Proper survey weighting would let the regional percentages mean what people think they mean. And a Jaccard-distance hierarchical clustering would give a similarity measure appropriate to binary indicator vectors, rather than Euclidean distance on a matrix where every point shares most of its zeros.

8 Academic honesty statement

I personally designed and carried out all data analysis procedures presented in this report. I wrote the report and created the figures myself. I used an LLM in the limited ways permitted by the course policy. For writing, I used it to brainstorm possible ways to improve wording and presentation and to suggest edits to grammar, wording, and sentence flow in text that I had already written. I did not use an LLM to generate paragraphs or produce report content from bullet points.

For coding, I used an LLM to brainstorm possible approaches and for suggestions related to parts of the modeling and sensitivity-check code, as well as figure revisions. I reviewed and understood the suggested code before using or adapting it and remained responsible for the final implementation and analysis.

I believe academic honesty is a fundamental requirement for conducting research. Researchers need to take responsibility for the results they produce because science builds on collaboration and on the work of others. If research is dishonest or cannot be reproduced, it can become the basis for further research and potentially lead to a chain of incorrect conclusions. Academic work should also be original and properly acknowledge the contributions of others. Plagiarism does not contribute new work to the scientific community and fails to give appropriate credit to the original authors. For these reasons, research should be conducted in an honest and reproducible way.

9 Collaborators

None.

Bibliography

- J. Nerbonne and W. Kretzschmar, Jr., “Introducing computational techniques in dialectometry,” *Computers and the Humanities* 37:245–255, 2003.
- J. Nerbonne, “Progress in dialectometry: toward explanation,” *Literary and Linguistic Computing*, 2006.
- B. Vaux, *Dialect Survey*, <https://www.dialectsofenglish.com/>.